

Cole Gruninger

Department of Mathematics, University of North Carolina at Chapel Hill
Email: coleco@live.unc.edu

Education

Ph.D. in Mathematics, University of North Carolina at Chapel Hill Expected August, 2025
Advisor: Professor Boyce E. Griffith and Mark Gregory Forest

B.S. Chemistry (Honors), B.A. Mathematics, University of North Carolina at Chapel Hill June, 2020

Research Interests

Numerical methods for solving partial differential equations; immersed boundary methods for fluid-structure interaction; viscoelastic and non-Newtonian fluids; electrochemical modeling.

Publications

Preprints

1. L. Li, C. Gruninger, J.H. Lee, B.E. Griffith, "Local Divergence-Free Immersed Finite Element-Difference Method Using Composite B-Splines." *arXiv preprint arXiv:2412.15408*, 2024.
2. C. Gruninger, B.E. Griffith, "Local divergence-free velocity interpolation for the immersed boundary method using composite B-splines." *arXiv preprint arXiv:2408.08280*, 2024.

Published

1. A. Barrett, A.L. Fogelson, M.G. Forest, C. Gruninger, S. Lim, and B.E. Griffith. "Flagellum pumping efficiency in shear-thinning viscoelastic fluids." *Journal of Fluid Mechanics*, 999:A2, 2024.
2. C. Gruninger, A. Barrett, F. Fang, MG Forest, BE Griffith, "Benchmarking the immersed boundary method for viscoelastic flows," *Journal of Computational Physics*, 506, 112888, 2024.
3. DP Isaacs, CT Gruninger, T. Huang, AM Jordan, G Nicholas, CH Chen, "Visible light induced formation of a tungsten hydride complex," *Dalton Transactions*, 52 (10), 3210-3218, 2023.
4. T. Huang, TM Rodriguez, CT Gruninger, DA Kurtz, AM Jordan, CH Chen, "Electrosynthetic route to cyclopentadienyl rhenium hydride complexes enabled by electrochemical investigations of their redox-induced formation," *Organometallics*, 39 (10), 1730-1743, 2020.
5. KJ Lee, KM Lodaya, CT Gruninger, ES Rountree, JL Dempsey, "Redox mediators accelerate electrochemically-driven solubility cycling of molecular transition metal complexes," *Chemical Science*, 11 (36), 9836-9851, 2020.
6. KJ Lee, CT Gruninger, KM Lodaya, S Qadeer, BE Griffith, JL Dempsey, "Analysis of multi-electron, multi-step homogeneous catalysis by rotating disc electrode voltammetry: theory, application, and obstacles," *Analyst*, 145 (4), 1258-1278, 2020.

Undergraduate Honors Thesis

C. Gruninger, "The Search For Photon-Induced Metal Hydride Formation and Modeling of Multi-Step Multi-Electron Electrocatalytic Reactions at the Rotating Disc Electrode," 2020.

Awards and Honors

National Defense Science and Engineering (NDSEG) Fellowship, Department of Defense, 2022.
James H. Maguire Memorial Award, Chemistry Department, University of North Carolina at Chapel Hill, 2019.
Summer of Undergraduate Research Fellowship, University of North Carolina at Chapel Hill, 2019.

Conference Talks

"Benchmarking the immersed boundary method for viscoelastic flows." SIAM Conference on Computational Science & Engineering. Fort Worth, Texas, U.S. March 3-7, 2025.

"Advancing the immersed boundary method: viscoelastic applications and volume conservation improvements." Colloquium, Department of Mathematics, Fordham University, Bronx, New York, 2025.

"Improving the volume conservation properties of the immersed boundary method using composite B-spline regularized delta functions." The 16th World Congress on Computational Mechanics and 4th Pan American Congress on Computational Mechanics. Vancouver, British Columbia, Canada. July 21-26, 2024.

"Benchmarking the immersed boundary method for viscoelastic flows and improving the volume conservation properties of the immersed boundary method using composite B-spline regularized delta functions." The Fifth Annual NDSEG Conference. New Orleans, July 14-18, 2024

"Benchmarking the immersed boundary method for viscoelastic flows." The AIMS Conference on Dynamical Systems, Differential Equations and Applications. UNC- Wilmington, May 31 - June 4, 2023

Poster Presentations

"Benchmarking the immersed boundary method for viscoelastic flows and improving the volume conservation properties of the immersed boundary method using composite B-spline regularized delta functions. Computational Tools for PDEs with Complicated Geometries and Interfaces. Flatiron Institute, June 10 - June 14, 2024.

Teaching Experience

Graduate Teaching Assistant, University of North Carolina at Chapel Hill, 2020 – 2021

Fall 2020: MATH 383. A First Course in Differential Equations

Spring 2021: MATH 381. Discrete Mathematics

Summer 2021: MATH 233. Calculus of Functions of Several Variables

Skills

Programming: Python, MATLAB, C++, LaTeX, Julia

References

Available upon request.

Website: <https://sites.google.com/view/colegruninger/about>